Orthographic Variation and Input Normalization in Persian LLM Multiple-Choice Question Answering: A Paired Pilot Protocol

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Abstract

Persian allows several spellings of the same word: the Arabic and Persian forms of yeh and kaf look identical on screen, and a word boundary may be a space, a zero-width non-joiner (ZWNJ) or nothing at all. Readers tolerate these variants, whereas a tokenizer processes them as different strings. We ask each multiple-choice question under six conditions: as written (C), with a controlled spelling change (P(C)), and after each of two normalizers. N is a deterministic inverse-map baseline, and H is Hazm 0.10.0. We approved the items by joint review, which left 97 yeh/kaf items and 69 ZWNJ-to-space items. Qwen2.5-1.5B-Instruct (Q4_K_M) answered three requests per eligible condition, 2,982 in all, and every response could be scored. On yeh/kaf, C and P(C) each got 138/291 requests and 46/97 item majorities right. Seven items got worse and seven got better (exact McNemar p=1.0), and the request-level difference was 0.0 points (95% bootstrap CI −7.2 to 7.2). On ZWNJ, C got 94/207 and P(C) 96/207, and both got 32/69 majorities. Two items got worse and two better (p=1.0), and the difference was +1.0 point (CI −3.9 to 5.8). N recovered 21 of 23 yeh/kaf request-level losses, all by restoring C exactly. H recovered 17 of 23, five of them without restoring C exactly. On ZWNJ, N recovered 1 of 5 without exact restoration, and H recovered 4 of 5 by exact restoration. Identical inputs still got different answers now and then. We saw no aggregate accuracy difference; because the intervals allow modest effects either way, this is no evidence of equivalence. The only model was small, quantized and limited in accuracy, so whether larger models behave differently is still open.

Keywords: Persian NLP; orthographic variation; input normalization; large language models; multiple-choice question answering; robustness.

Summary. Item-majority accuracy under C and P(C) was identical in both families (46/97 for yeh/kaf, 32/69 for ZWNJ). The results come from a single small quantized model that answers fewer than half of the questions correctly, and the confidence intervals allow modest effects in either direction. For this model and these items we therefore observed no aggregate difference; the data do not show that spelling is irrelevant for Persian LLMs.

1. Introduction

A Persian question has no single canonical spelling: the Arabic and Persian forms of yeh (ى ,ي) and kaf (ک ,ك) are different code points that most readers cannot tell apart, and word boundaries are just as loose, since an ordinary space, a ZWNJ or no visible gap can all occur in the same position. Readers are largely unaffected by this variation; the tokenizer, however, receives different strings and the model different token sequences.

We already know that small prompt changes can move model answers. Romanou et al. [1] show it on several established benchmarks and separate how hard an item is from how sensitive it is to its surface form. Aliakbarzadeh et al. [2] build noisy multilingual test sets from Wikipedia edit histories; Persian is not among their languages. For Persian itself, normalization has been studied in sentiment classification [3], colloquial-to-standard rewriting [4] and retrieval [5]. None of these follows answer changes through paired Persian multiple-choice questions, and that is the gap this protocol goes after.

We ask whether a spelling change that a reader would not notice alters the model's answer, and whether a fixed normalizer then restores it or breaks an answer that was correct. Normalization that simply returns the text to its original bytes will of course look like a perfect fix, so those cases get reported on their own.

2. Related Work

The cited studies examine different interventions, so their numbers are not pooled; we group them by their relevance to this protocol.

2.1 Prompt sensitivity and noise

Brittlebench [1] applies word-level, formatting and paraphrase changes to several benchmarks and compares each item with its own perturbed versions. We take that within-item logic directly. Brittlebench includes neither Persian Unicode variants nor a step in which a normalizer repairs the damage. Aliakbarzadeh et al. [2] draw noisy test sets for NLI, NER and intent classification in six languages from Wikipedia edit histories. Their comparison of clean and noisy versions covers no Persian, no multiple-choice QA and no normalization.

2.2 Robustness beyond spelling

Bogavelli et al. [6] test punctuation, whitespace, structured formats, multilingual inputs and instruction position in enterprise settings, and report losses of up to 40 percentage points on some metrics across 11 models. Their figure is specific to their setting, yet it gives a reason to measure "minor" edits instead of assuming they are harmless. Nhayakkat et al. [7] reorder constituents and flip active and passive in Hindi and Malayalam math problems, and they treat meaning preservation as something the experiment has to check. ObfusQAte [12] goes another way, using entity indirection, distractors and contextual overload on factual QA. Although none of this concerns Unicode normalization, it shows why a robustness study should state exactly which perturbations it uses.

Sperduti and Moreo [8] survey the wider misspelling literature. They split noise into natural, artificial, hybrid and adversarial, point out that no shared robustness benchmark exists, and review evidence that synthetic misspellings can behave differently from the errors people really make. So we report the exact character map, the edit positions and the number of changed characters rather than calling everything "noise".

2.3 Persian orthography and normalization

Khazeni et al. [3] describe a slang-to-formal pipeline whose preprocessing unifies Persian and Arabic letters and fixes spaces and semi-spaces. Ghayoomi et al. [4] test five web LLMs on colloquial-to-standard rewriting and sort the changes against a Persian orthography guide. Naebzadeh et al. [10] survey the informal-to-formal area and point to gaps in length balance and dialect coverage. Zoughi et al. [5] handle character and whitespace

normalization inside a Persian RAG system. They all show the problem is real. What they measure is rewriting quality or downstream classification, not whether an answer survives a spelling change.

Dashti et al. [11] bring in context. Their spelling corrector, PERCORE, combines contextual representations with phonetic analysis and scores real-word and non-word errors separately. A character map can be exactly reversible in Unicode and still alter the meaning of a particular question, so we screen out spelling-sensitive items and review each ZWNJ edit individually.

Abdullah et al. [9] work on the representation side. They train Arabic-script RoBERTa models separately for Kurdish Sorani, Arabic, Persian and Urdu, and report gains of 2–5 percentage points over multilingual baselines on language-specific classification. That supports handling shared-script variation language by language; it does not tell us what an off-the-shelf LLM does with a yeh or kaf swap. Two more papers are here only to mark the edge of the term: Chen et al. [20] study normalization inside transformer architectures, and Sunkara et al. [21] study inverse text normalization for speech recognition. Neither one deals with Persian input.

2.4 Benchmark quality and human review

The FarsEval-PKBETS paper [26] describes a 4,000-question Persian benchmark across several response formats and domains, with two main reviewers who supervised question quality and sent items back for revision. It is our best example of why human verification matters. We haven't matched our five local JSON files to that release, so we cite the paper as a benchmark source and keep the files under their repository name [25].

Other benchmarks make the same point. MedHopQA [18] has 1,000 expert-curated biomedical questions validated through a human–AI pipeline. ToolQA [19] leans on automated curation and tool-based verification. PerSoMed [17] pairs ChatGPT few-shot annotation with human checking and gets weaker results on the more ambiguous social and political classes. None of this validates our items, although it shows that the reviewer, the protocol and the disagreements should all be written down.

Liu [24] compares gpt-4o-mini labels with three crowdworker labels on a rubric task and finds that disagreements cluster around adjacent severity levels. Although it is a subjective-labeling study rather than factual MCQ validation, it supports our decision to leave the human-verification fields empty after the AI-assisted first pass. The eight flagged cultural and pragmatic items are where a Persian-speaking reviewer is most likely to disagree with a quick screen. Mortazavi et al. [14] call Persian annotation-scarce: uneven coverage across tasks and domains, patchy documentation and little specialist supervision. A dataset can hold many records and still lack the item-level evidence a causal claim needs, so the 144/8/5 split below is data preparation, not a quality assessment.

2.5 Persian models and resources

Hakim [15] is a Persian embedding model released with new retrieval datasets. Persian-Phi [16] adapts the compact 3.8B Phi model through continued pretraining and parameter-efficient tuning, and PersianLLaMA [23] reports 7B and 13B Persian models trained on formal and colloquial text. Although they are relevant to model choice on a small budget, none of them runs an experiment on orthographic pairs. ROBUSTFT [13] deals with noisy responses during supervised fine-tuning, which is a training-data problem, not input noise. The Persian proverb study [22] builds a culturally grounded story dataset with human-calibrated evaluation and shows how hard cultural meaning stays even when the text is fluent. Both are reasons to keep culture-sensitive questions visible in the review record.

2.6 Where this leaves the study

Prompt sensitivity, structural perturbation, misspelling robustness, Persian standardization and spelling correction, Persian benchmarks and human–AI annotation are all covered somewhere. What nobody has run, as far as our collection shows, is this: on author-approved Persian MCQs, do reversible Arabic/Persian character or spacing changes alter answer correctness, and do fixed normalization procedures bring the reference answer back without adding errors? This study addresses that gap.

3. Research Questions

1. **RQ1 (Sensitivity).** Among author-approved Persian MCQs, how often does a controlled, meaning-preserving orthographic variant turn a correct answer into an incorrect one?
2. **RQ2 (Recovery and harm).** Among items answered correctly in C and incorrectly in P(C), how often are N(P(C)) and H(P(C)) correct? How often does either normalizer turn a correct answer into a wrong one?
3. **RQ3 (Interpretation and repeatability).** How do exact-restoration cases differ from the rest, and how often do repeated identical inputs get different labels?

4. Data and Item Screening

4.1 The local files

The five local files come from the main/data directory of Persian LLM Eval; we didn't record the exact commit. They hold 554 records: 100 challenge, 10 development, 145 hard, 150 practical and 149 public-evaluation. No identifier appears twice. Of these, 157 are marked as multiple-choice, and in every one of them the answer index points to the same text as the stored answer. The metadata tag 250 records as *pending_review* and 254 as *accepted*, and 50 carry no review status at all. Those tags describe the files; they don't tell us anyone checked the answers.

The two of us, both Persian speakers, then reviewed the 40 pilot questions, their keyed answers and whether meaning survived in all four conditions. We worked independently first and settled disagreements together. This is human verification done by the authors, not an outside review. We agreed that every yeh/kaf variant kept the intended meaning. We excluded `peval-hard-reasoning-002` and `peval-hard-reasoning-024`, and we classed `peval-hard-reasoning-005`, `-007` and `-012` as ambiguous and left them out of the cleaned analysis.

After the ZWNJ run, we reviewed all 24 ZWNJ-eligible questions, their choices, stored keys and transformed texts, again working independently, and our decisions were identical. Twenty items were approved for a post hoc reviewed analysis. We rejected reasoning-005 and reasoning-024 for key or inference defects. We excluded reasoning-009 as uncertain, because its conditional states an intention and not a guaranteed rule. reasoning-007 stayed out under the earlier ambiguity decision. We also checked that all 11 clean-input edits made by Hazm preserved meaning and the stored answer. As before, this is author-conducted review, not external expert validation.

On 21 October 2026 we jointly reviewed the full text and four choices of the 117 structurally consistent MCQs outside the 40-item pilot. This was joint adjudication, not independent rating, so we make no kappa or agreement claim. We approved 97 items, marked 11 for rewriting, rejected two semantic duplicates and left seven uncertain. Then we jointly reviewed every generated expansion transformation before anything ran. All 97 yeh/kaf P/N branches were approved. Only 96 were approved for all six conditions, because Hazm changed the meaning of `peval-public-knowledge-002`. All six conditions were approved for the 69 ZWNJ-eligible items. These decisions defined the frozen expansion sets.

Our paper collection holds 26 PDFs but only 25 distinct files, because 2504.14690v1.pdf and V11.pdf are byte-identical.

4.2 Preliminary screening

All 157 questions went through a quick editorial screen done with AI assistance, not by an independent human reviewer. For each item it compared the stem, the choices and the stored answer index, and it looked for duplicate answers, ambiguous wording, missing choices, a mismatch between question and choices, and answers that depend on time, region or culture. It did not check facts against outside sources. Table 1 and Figure 1 show the outcome.

Outcome	Items
Usable for now	144
Needs a human check	8
Exclude or rewrite	5
Total screened	**157**

Table 1. Result of the preliminary screening.



Figure 1. Preliminary screening of the 157 structurally consistent items (n = 157). The chart describes data preparation only; it is not a model-accuracy result.

The 144 items showed no obvious competing answer or structural defect, which is a lower bar than having verified facts. The eight held back involve culturally variable expressions, folk beliefs, context-dependent pragmatics or knowledge that changes over time. Five items have a plain defect and stay out of the pilot as written (Table 2). Table 3 gives all thirteen flagged items with their status after our later joint review, translated from the original Persian notes. That review didn't affect any completed model run. The joint record of the 117 non-pilot items stays under `work/` because it contains third-party text.

Record ID	Reason for exclusion or revision
peval-hard-reasoning-013	Two choices give effectively the same answer, so the keyed answer is not unique.
peval-hard-culture-020	The expression جا انداختن is ambiguous without more context.
peval-public-knowledge-014	The stem asks for an element and its periodic-table group, but the choices are element names only.
peval-public-knowledge-017	The stem asks for a protocol and its security layer, but no TLS/SSL choice is offered.
peval-public-knowledge-023	The question concerns a section of the Shahnameh, while the choices are names of poets.

Table 2. Items marked for exclusion or rewriting.

Record ID	Status	Rationale	Example or implication
peval-hard-reasoning-013	Rewrite	Two choices express effectively the same answer.	Our joint answer was A|D; replace option D.
peval-hard-culture-020	Rewrite	جا انداختن has several meanings without enough context.	Add explicit classroom context or remove the item.
peval-public-knowledge-014	Rewrite	The stem asks for an element and group, but the choices give only names.	Ask only for the element, or add groups.
peval-public-knowledge-017	Rewrite	The HTTPS/TLS answer is absent.	We recorded NONE.
peval-public-knowledge-023	Rewrite	The stem and the answer type don't match.	We recorded NONE.
peval-hard-culture-001	Uncertain	The idiom depends on context, and the keyed gloss is broad.	Needs review by people from more varied backgrounds.
peval-hard-culture-002	Uncertain	Rejecting taarof depends on context.	The keyed choice is plausible but weakly constrained.
peval-hard-culture-009	Rewrite	Calling the utterance "polite" is contestable.	Use "a direct request not to interfere."
peval-hard-culture-015	Rewrite	No universal repetition count exists for taarof.	Remove the arbitrary "one or two" wording.
peval-practical-pragmatics-006	Uncertain	صبر آمد varies by region and generation.	Needs review by people from more varied backgrounds.
peval-public-knowledge-005	Rewrite	The population ranking changes over time.	Add a census year and recheck the key.
peval-public-culture-016	Rewrite	The keyed choice combines two occasions.	Separate the occasions or narrow the stem.
peval-public-culture-019	Uncertain	The expression depends on context.	Drinking is plausible, but "most often" is contestable.

Table 3. The thirteen initially flagged items after joint review. The defects in public-knowledge-014 and -017 are structural, because no option can fully answer the stem. Items like hard-culture-002 and public-culture-019 may become answerable once the social context is stated. We therefore keep "rewrite" and "uncertain" as separate categories.

The author review covered only the 40 pilot items. The rest of the 144 usable questions and the 8 flagged ones still need the same treatment: confirm there is one defensible answer, drop anything ambiguous or sensitive to spelling, and check that each variant keeps the intended meaning. The review sheets list record IDs and SHA-256 prompt hashes instead of question text, since we haven't settled whether the data can be redistributed. The human-verification columns in the screening CSV stay empty.

5. Experimental Design

5.1 Six paired conditions

Each question is asked under up to six conditions (Table 4, Figure 2). Choices, labels, order and the response instruction are byte-identical in all of them, and the model never sees the reference answer. The original four-condition output stays the default. The flag `--hazm-normalizer` adds the two Hazm conditions explicitly, and the machine labels stay backward compatible: `NC`, `PC`, `NPC`, `HC` and `HPC`.

Condition	Description	Model-visible change
C	Verified reference	None
P(C)	Controlled variant	Question body only
N(C)	C after our explicit rule-based normalizer	Custom rules
N(P(C))	P(C) after the same normalizer	Custom rules
H(C)	C after Hazm 0.10.0	Off-the-shelf normalizer
H(P(C))	P(C) after Hazm 0.10.0	Off-the-shelf normalizer

Table 4. The six paired conditions.

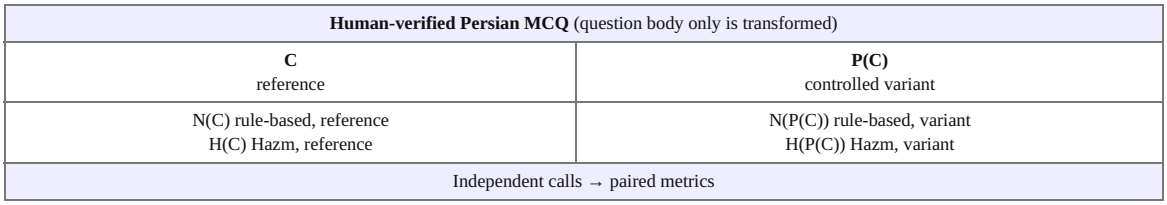


Figure 2. Study design. Each question branches into six conditions, each is sent in an independent call, and the results are analyzed as pairs. The figure shows the design only.

5.2 Perturbations

The character perturbation swaps Persian yeh and kaf for their Arabic counterparts: ي → ی and ك → ک. N applies that reverse map explicitly, plus only the literal spacing rules. The spacing perturbation replaces an existing ZWNJ with an ordinary space and logs every edit. Its rules live in `code/spacing_rules.json`, and nothing deletes spaces globally. If a transformation leaves the string unchanged, the variant is thrown away.

Although ZWNJ edits look harmless in most words, they can change how a tokenizer splits a word and occasionally what the word means, so each one gets an item-level review. The yeh/kaf swap differs in that it can be undone exactly, so N(P(C)) is often byte-identical to C. We analyze those exact-restoration cases separately so they don't inflate the recovery rate.

Hazm is a more demanding test of the normalization step. H is `hazm.Normalizer` from Hazm 0.10.0 [28] with all eight constructor switches on: `correct_spacing`, `remove_diacritics`, `remove_specials_chars`, `decrease_repeated_chars`, `persian_style`, `persian_numbers`, `unicodes_replacement` and `seperate_mi`. It does more than undo one character swap, so it can change a clean question as well. Every textual edit it makes is logged. That makes the comparison with N meaningful: a Hazm "recovery" is no longer just the original string coming back.

5.3 Response format

Every condition runs in a fresh context with the same short instruction, which asks for the letter of the correct choice and nothing else:

فقط حرف گزینهٔ درست را بنویس (مثلاً A). هیچ توضیح دیگری ننویس.

The code prepares requests offline, imports responses produced elsewhere and resumes a capped run. It calls an API only with explicit authorization and the `--execute` flag.

5.4 The pilot runs

We ran four local experiments with the same model and engine. The model was `Qwen2.5-1.5B-Instruct-Q4_K_M.gguf` (SHA-256 `1adf0b11065d8ad2e8123ea110d1ec956dab4ab038eab665614adba04b6c3370`), served through llama.cpp build 11612 (commit 10da8c463). The model repository declares Apache-2.0 terms [27]. Inference ran on an Intel i5-6500 CPU with four logical processors and about 12 GB of RAM. No paid API was involved, and every condition went to the local server as its own request.

The first pilot, on 17 October 2026, used four conditions on 40 items from the screened pool, one request each. The model ran before any human check: the reference answers came from the AI-assisted screen, and the author review in Section 4.1 came afterwards. It used temperature 0, seed 20261017, a one-token cap, a grammar limiting output to A–D, and shuffled request order.

The second run, on 18 October 2026, is the one we analyze. It kept the model, engine, decoding settings and shuffle seed, added H(C) and H(P(C)), and sent three independent requests per item and condition. It tested only the yeh/kaf swap, applied to the question body and not to the choices, with the spacing rules switched off. All 720 responses could be scored, and no request failed. Hazm 0.10.0 pins NumPy 1.24.3, which doesn't work with Python 3.12, so the run used Hazm 0.10.0 with documented dependency overrides.

The third run tested the ZWNJ-to-space change on the 24 eligible items, with the same model settings, all six conditions and three repeats: 432 requests, all scorable, none failed. It ran before our detailed two-author review of the ZWNJ variants.

The expansion ran on 21 October 2026 with the same model, engine, prompt, decoding constraints, seed and three repeats. For yeh/kaf it sent 1,164 requests for C, P(C), N(C) and N(P(C)) on 97 items, and 576 requests on the 96-item subset that was safe for Hazm. For ZWNJ it sent 1,242 requests across all six conditions on 69 items. All 2,982 responses were scorable, and none failed in transit.

6. Analysis Plan

We wrote the analysis down before the six-condition run, in `analysis_plan_v8.md`. It fixed the 35-item set kept by the author review as the primary analysis and all 40 items as a secondary one, with three requests per item and condition. The five exclusions come from our earlier review (Section 4.1), so they were settled before this run and not after seeing its results. The plan named these quantities: request-level and item-majority accuracy, perturbation loss among C-correct observations, N-versus-H recovery, harm on clean and on perturbed inputs, exact recovery separated from non-exact recovery, and disagreement between identical inputs. Anything we add beyond that is labelled post hoc.

Before the ZWNJ run we wrote a second plan. It fixed all 24 eligible items as a transparency set and 21 items as the exploratory primary set, dropping three records already flagged for item-quality concerns: `peval-hard-reasoning-005`, `-007` and `-024`. Our detailed review of the ZWNJ variants came after the run. The 20-item set that also drops the uncertain `-009` is therefore a post hoc sensitivity analysis. It doesn't replace the frozen 21-item result.

Accuracy is reported for every condition that was run. Whether H(P(C)) equals H(C) is a different question from whether H(P(C)) equals the original C, and we keep the two apart. Transport failures and unscorable outputs are counted separately from wrong answers. A zero denominator is stored as *null*, not as zero.

We don't treat N as a rival to H. For yeh/kaf, N is a deterministic inverse-map positive control, and its exact recoveries are expected by construction. H is a general-purpose, off-the-shelf normalizer that may change more than we asked of it. So we report their recovery, harm and exact-restoration counts separately and make no claim about which one ranks higher.

For the expansion, the primary inferential endpoint was item-majority correctness for C versus P(C), fixed in advance. We use a two-sided exact McNemar test with one paired majority outcome per item. For request-level differences between C and P(C), we give percentile 95% confidence intervals from 10,000 item-clustered bootstrap samples (seed 20261021), keeping all repeats of each sampled item together. Identical inputs still vary between calls. We keep that as a stated limit and still run the paired analysis.

The earlier pilot runs (40 and 24 items) got no confidence intervals or tests. Their denominators are small, and five items were filtered only after the first run.

7. Results

The expansion completed all 2,982 requests, and every response could be scored. We don't pool the two perturbation families.

7.1 Yeh/kaf expansion

Condition	Request accuracy	Item-majority accuracy
C	138/291 (47.4%)	46/97 (47.4%)
P(C)	138/291 (47.4%)	46/97 (47.4%)
N(C)	138/291 (47.4%)	45/97 (46.4%)
N(P(C))	141/291 (48.5%)	48/97 (49.5%)
H(C)	134/288 (46.5%)	45/96 (46.9%)
H(P(C))	133/288 (46.2%)	42/96 (43.8%)

Table 5. Yeh/kaf expansion accuracy. The Hazm rows cover the 96-item subset.

For C versus P(C), 39 items were right in both conditions and 44 wrong in both. Seven went from right to wrong and seven from wrong to right. Item-majority accuracy didn't move, and the exact McNemar result was p=1.0. At request level the difference was 0.0 points, with a 95% CI from −7.2 to 7.2.

7.2 ZWNJ expansion

Condition	Request accuracy	Item-majority accuracy
C	94/207 (45.4%)	32/69 (46.4%)
P(C)	96/207 (46.4%)	32/69 (46.4%)
N(C)	95/207 (45.9%)	32/69 (46.4%)
N(P(C))	95/207 (45.9%)	32/69 (46.4%)
H(C)	91/207 (44.0%)	30/69 (43.5%)
H(P(C))	94/207 (45.4%)	31/69 (44.9%)

Table 6. ZWNJ expansion accuracy.

For C versus P(C), 30 items were right in both and 35 wrong in both. Two got worse and two got better. Item-majority accuracy was identical (exact McNemar p=1.0), and the request-level difference P(C)−C was +1.0 point (95% CI −3.9 to 5.8).

7.3 Recovery and harm in the expansion

Metric	Yeh/kaf, N	Yeh/kaf, H	ZWNJ, N	ZWNJ, H
Perturbation loss	23/138 C-correct requests		5/94 C-correct requests	
Recovery among losses	21/23	17/23	1/5	4/5
Exact recovery (text equals C)	21/23	12/23	0/5	4/5
Non-exact recovery	0/23	5/23	1/5	0/5
Harm on clean input	8/138	8/135	1/94	4/94
Harm on perturbed input	22/138	23/135	5/96	6/96

Table 7. Recovery and harm in the expansion, by perturbation family.

On yeh/kaf, every N recovery restored C exactly, which is what an inverse map does by construction. H recovered 17 of the 23 losses: 12 by restoring C exactly and five without. N(C) and N(P(C)) were the same string as C in this family, so differences between their independent calls are the model varying, not changed text. Among 214 repeated identical-input groups, 22 disagreed in label and 19 in correctness.

On ZWNJ, N recovered one loss without restoring C, and H recovered four by restoring C exactly. Sixteen of 174 repeated identical-input groups disagreed in label and ten in correctness. These counts don't support a general ZWNJ accuracy effect or a general benefit from either normalizer.

7.4 Earlier yeh/kaf run (40 items)

The first six-condition run completed 720 of 720 requests. Its 35-item analysis was fixed in advance as primary, and all 40 items as secondary. The similarly filtered Version 7 result is post hoc, and we don't pool it here.

Condition	Primary requests (n=105)	Primary majority (n=35)	Secondary requests (n=120)	Secondary majority (n=40)
C	53/105 (50.5%)	17/35 (48.6%)	54/120 (45.0%)	17/40 (42.5%)
P(C)	52/105 (49.5%)	18/35 (51.4%)	53/120 (44.2%)	18/40 (45.0%)
N(C)	52/105 (49.5%)	17/35 (48.6%)	53/120 (44.2%)	17/40 (42.5%)
N(P(C))	52/105 (49.5%)	17/35 (48.6%)	53/120 (44.2%)	17/40 (42.5%)
H(C)	52/105 (49.5%)	17/35 (48.6%)	55/120 (45.8%)	18/40 (45.0%)
H(P(C))	51/105 (48.6%)	17/35 (48.6%)	52/120 (43.3%)	17/40 (42.5%)

Table 8. Accuracy in the first six-condition yeh/kaf run.

Primary paired metric	N	H
Perturbation loss	6/53 C-correct observations (11.3%)	
Recovery among losses	5/6	5/6
Exact recovery	5/6	5/6
Non-exact recovery	0/6	0/6
Harm on clean input	3/53	2/53
Harm on perturbed input	5/52	6/52

Table 9. Paired metrics in the first six-condition yeh/kaf run.

N(C) was the same string as C, so N's three clean losses are the model varying between calls. Although both of H's clean losses involved text Hazm had really changed, independent calls and denominators this small don't let us call that causal. Fourteen of 79 primary identical-input groups disagreed in label, and six in correctness. Every recovery exactly recreated C.

7.5 Earlier ZWNJ run (24 items)

The first ZWNJ run completed 432 of 432 requests. On its frozen 21-item primary set, request accuracy was 27/63 (42.9%) for C and P(C), 28/63 (44.4%) for N(C), 26/63 (41.3%) for N(P(C)) and 28/63 (44.4%) for both Hazm conditions. Majority accuracy was 9/21 everywhere except N(P(C)), at 8/21.

Primary paired metric	N	H
Perturbation loss	3/27 C-correct observations	
Recovery among losses	1/3	3/3
Exact recovery	0/3	3/3
Non-exact recovery	1/3	0/3
Harm on clean input	0/27	0/27
Harm on perturbed input	2/27	2/27

Table 10. Paired metrics in the first ZWNJ run, frozen 21-item primary set.

All three losses came from one item, `peval-hard-reasoning-019`, whose stored answer was C ("two people"). Across three repeats, C answered C, C, C, P(C) answered A, A, A, N(P(C)) answered C, A, A and H(P(C)) answered C, C, C. The perturbation replaced three instances of میدانند with می دانند. Hazm restored C exactly, and N did not. The item repeated across calls. That is an item-level observation and no evidence of a general ZWNJ effect.

Both perturbed-input harms in the ZWNJ run sit on `peval-challenge-ambiguity-016`. There N(P(C)) was text-identical to P(C), and H(P(C)) was text-identical to C and H(C), so the answer differences are the model varying between identical inputs. Four of 50 repeated identical-input groups disagreed in label, and three in correctness.

After the run, we each returned identical review decisions. The post hoc 20-item set drops reasoning-005, -007, -009 and -024. Its request accuracies were 27/60 (45.0%) for C and P(C), 28/60 (46.7%) for N(C), 26/60 (43.3%) for N(P(C)), and 28/60 (46.7%) for both Hazm conditions. Loss and recovery counts didn't change.

7.6 A closer look at reasoning-019

This is a post hoc analysis. The item is a short set-overlap problem. With seven people, at least four French speakers and at least five English speakers, the smallest possible overlap is 4 + 5 − 7 = 2, which is option C. Both of us confirmed that the clean and the perturbed wording keep this reasoning and the single correct answer. All three edits fall in the repeated verb میدانند. No number, quantifier, option or logical relation changes.

We tokenized both versions with the exact Qwen2.5 model file from the experiment. Both came to 76 tokens, so a longer prompt doesn't explain the effect. Each edited verb did change its local token boundaries, though: the ZWNJ-bearing form and the space-bearing form of میدانند split into different token sequences. So the intervention swapped three ZWNJ-bearing token sequences for three different space-bearing ones and left the token count alone.

One possible reading is that repeated local changes in representation unsettled a small quantized model on a multi-step arithmetic item. That is a hypothesis we formed after the run. We didn't inspect internal activations, only one item showed the full pattern of C right and P(C) wrong, and identical inputs varied elsewhere. A larger preregistered study should test whether sensitivity grows with the number of altered ZWNJs, with verb-prefix edits, or with the number of reasoning steps.

7.7 Every stage of the project

Stage	Items and conditions	Requests	Role here
Initial feasibility pilot	5 items × 4 conditions	20	Historical check
Expanded one-repeat pilot	40 items × 4 conditions	160	Historical, not pooled
Repeated yeh/kaf run	40 items × 6 conditions × 3 repeats	720	Analysis fixed in advance
Repeated ZWNJ run	24 items × 6 conditions × 3 repeats	432	Later exploratory run; reviewed-set filtering is post hoc
Yeh/kaf expansion	97 P/N-safe items, 96 Hazm-safe items, 3 repeats	1,740	Plan frozen before execution
ZWNJ expansion	69 items × 6 conditions × 3 repeats	1,242	Plan frozen before execution

Table 11. Execution record. Every run was local and used no paid API. We show counts by stage and don't add them up, since the stages share items and conditions.

7.8 Text-level checks

For both perturbation families, all six conditions were generated before the corresponding model run.

Text-level check	Yeh/kaf	ZWNJ to space
Eligible items	40	24
H(C) = H(P(C))	40/40	20/24
H(P(C)) = C	29/40	18/24
N(P(C)) = C	37/40 (spacing rules on)	0/24

Table 12. Deterministic comparison of the two normalizers on the first 40 and 24 items. These are text results only.

For yeh/kaf, H(C) and H(P(C)) were identical in all 40 pairs, but H(P(C)) equalled C in only 29, because Hazm also changed clean text. For ZWNJ, Hazm restored C exactly in 18 of 24 pairs, and N in none, since N follows only its literal spacing rules.

7.9 What Hazm did to the clean questions

Hazm changed 11 of the 40 clean questions through 18 edit spans: 16 space-to-ZWNJ replacements and two deletions of fathatan (U+064B). After the run, both of us independently reviewed all 11 and agreed that meaning and the stored answer were preserved. That review is post hoc. Exact spans and longer notes are in `hazm_clean_change_audit.md`.

Record	Hazm change (original → Hazm)	Category	Our assessment
ambiguity-006	واقعاً ← واقعا	diacritic removal	preserved
ambiguity-008	گم کرده ← گمکرده	verb spacing	preserved
dev-culture-001	هفت سین ← هفتسین؛ سین های ← سینهای	compound and suffix	preserved
dev-culture-002	می شود ← میشود	prefix verb	preserved
dev-knowledge-002	ماه های ← ماههای	suffix plural	preserved
hard-culture-004	مناسب تر ← مناسبتر	suffix comparative	preserved
reasoning-001	three × می شود ← میشود	verb prefix	preserved
reasoning-002	محرمانه ای ← محرمانهای؛ می آید ← میآید	suffix and verb prefix	preserved; item already excluded for another defect
reasoning-004	four verb-prefix spacing repairs	verb prefix	preserved
reasoning-007	دقیقاً ← دقیقا	diacritic removal	preserved; item already marked ambiguous
reasoning-010	شطرنج بازی ← شطرنجبازی	compound and verb spacing	preserved; confirmed explicitly by both of us

Table 13. Hazm changes to the 11 clean questions it altered.

8. Conclusion

The expansion sent 2,982 local requests at no cost, on 97 yeh/kaf and 69 ZWNJ items. C and P(C) had identical item-majority accuracy in both families. The confidence intervals leave room for modest effects in either direction, and individual answers did change under both perturbations. N's yeh/kaf recoveries were exact restoration by construction, while H produced both exact and non-exact ones. Neither pattern shows a general accuracy benefit.

The study provides a reproducible paired workflow and evidence that individual answers can flip under both perturbations. It cannot establish that spelling is irrelevant, that the two conditions are statistically equivalent, or that Persian LLMs are robust in general. The only model was small, quantized and right on fewer than half the questions, which makes a null aggregate result less informative than it looks. A larger or more capable model could behave differently. We also approved the items ourselves, with no outside reviewer, so the 97 and 69 items are author-approved and not externally validated. The next step is to replicate with at least one substantially more capable model, then to bring in a reviewer from outside the author team.

9. Data and Code Availability

The package contains this manuscript and its HTML version, the frozen pilot and expansion analysis plans, the aggregate yeh/kaf and ZWNJ metrics, summary-level records of our review decisions, the Hazm change audit, the references, the literature and novelty audits, a Python pipeline that prepares all six conditions, 26 passing tests, the pinned local library versions and the design figures. Hazm is not bundled. Raw data, PDFs, model weights, runtime binaries, item-level prompts, raw responses and API keys are left out. The item-level review files and the run logs stay under `work/`, because we haven't settled whether the data can be redistributed.

10. Declarations

Author order, the affiliation shown above and co-corresponding authorship are confirmed. Funding, conflicts of interest, individual contributions and ethics applicability are still open. AI-assisted tools were used for drafting, code support and the preliminary screening described in Section 4.2. Both authors reviewed the study materials, methods, analysis and final manuscript and remain responsible for the work. [Disclosure wording to be adjusted to the target venue's policy.]

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Appendix: Reproducing the Pilot

```
.\work\python\python.exe output\code\audit_data.py --data-dir data --output-dir output
.\work\python\python.exe output\code\preliminary_review.py
.\work\python\python.exe -m unittest discover -s output\tests -v
.\work\python\python.exe output\code\pipeline.py prepare --allow-unverified `
  --record-ids-file work\pilot_2026-10-17_40\selected_record_ids.txt `
  --spacing-rules work\pilot_2026-10-17_40\no_spacing_rules.json `
  --variants arabic_yeh_kaf --repeats 1 --shuffle-seed 20261017
```

The `--allow-unverified` flag records that the model ran before our two-author review. The completed localhost run added `--single-label-grammar --seed 20261017`. Raw response logs stay local.

The executed six-condition yeh/kaf run used Hazm 0.10.0, three repeats and the explicit opt-in flag:

```
.\work\python\python.exe output\code\pipeline.py prepare `
  --allow-unverified --hazm-normalizer `
  --record-ids-file work\pilot_2026-10-17_40\selected_record_ids.txt `
  --variants arabic_yeh_kaf --repeats 3 --shuffle-seed 20261017 `
  --spacing-rules work\pilot_2026-10-17_40\no_spacing_rules.json `
  --conditions-out work\pilot_2026-10-18_v8_run\conditions.jsonl `
  --requests-out work\pilot_2026-10-18_v8_run\requests.jsonl
```

The later ZWNJ run used the same local model settings and three repeats:

```
.\work\python\python.exe output\code\pipeline.py prepare `
  --allow-unverified --hazm-normalizer `
  --record-ids-file work\pilot_2026-10-20_zwnj_run\record_ids.txt `
  --variants zwnj_to_space --repeats 3 --shuffle-seed 20261017 `
  --spacing-rules output\code\spacing_rules.json `
  --conditions-out work\pilot_2026-10-20_zwnj_run\conditions.jsonl `
  --requests-out work\pilot_2026-10-20_zwnj_run\requests.jsonl
```

Our detailed review of the ZWNJ variants came after execution, so the 20-item reviewed analysis is labelled post hoc throughout.

For the expansion, the joint review and `ANALYSIS_PLAN_EXPANSION_V12.md` were finished before anything ran. We split yeh/kaf into a 97-item four-condition run and a 96-item Hazm run, so the one question whose meaning Hazm changed (`public-knowledge-002`) never reached the model. The ZWNJ run used all six conditions on 69 items. Raw logs stay local.

Submission preview · one-model results; no larger-model replication yet.